CSP-203: Software Tools & Technologies Lab

Chapter – 3

Document Processing: LaTeX (Lecture – 1)

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Latex Outline

- Latex Introduction
- Document class – options – environments
- Itemize – enumerate
- Math
- Getting started with Overleaf [Demo]

What is LATEX?

- ***Typesetting:*** is the composition of text by means of arranging physical *type* (or *sort*) in mechanical systems

- ***Typesetter:*** *a person who typesets the text*

- ***TeX:*** is a typesetting system → designed and written → Prof. Donald Knuth and first released in 1978.

- ***LaTeX: (program)*** is a special version of TeX → created by → Prof. Leslie Lamport and first released in 1984.





LaTeX

The LaTeX Project

Original author(s)	Leslie Lamport
Initial release	1984; 39 years ago
Stable release	June 2023 LaTeX release[1] / June 2023; 3 months ago
Repository	github.com/latex3/latex2e
Type	Typesetting
License	LaTeX Project Public License (LPPL)
Website	latex-project.org

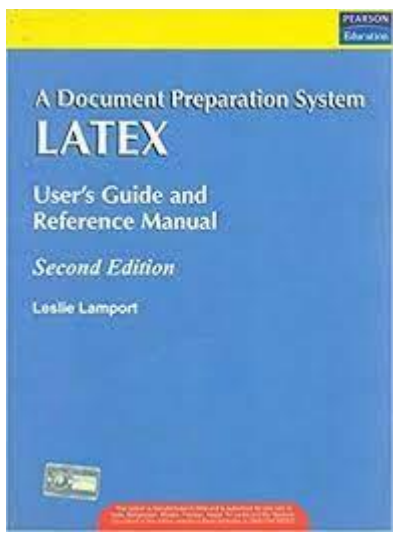
TeX

TEX

Developer(s)	Donald Knuth
Initial release	1978; 45 years ago
Stable release	3.141592653 / February 2021; 2 years ago
Repository	www.tug.org/svn/texlive/
Written in	WEB/Pascal
Operating system	Cross-platform
Type	Typesetting
License	Permissive free software
Website	tug.org

TeX

Filename extension	.tex
Internet media type	application/x-tex [a]
Initial release	1978; 45 years ago
Type of format	Document file format



Why LATEX?

- ***WYSIWYG*** → *What You See Is What You Get*



Pros

- Free!
- Looks pretty, especially math
- Consistent formatting
  - Good for long documents, such as dissertation and books
- Supporting ecosystem
  - Most journals have their LaTeX *style* files
- Good bibliography management
  - Bibtex supported widely

Cons

- Steep learning curve
- Hard to learn, but powerful
- *Can* meet most user requirements
  - but tricky to find out *how*
- Not WYSIWYG
  - Plain text with markup vs. Formatted text

LATEX : Components

hello.tex

```
\document class{article}  → Component 1
\begin{document}  → Component 2
Hello World! % body of the content
\end{document}  → Component 3
```

Offline mode

- Download and install
  - Backend: MikTeX
  - Frontend: Editor
    - TexStudio, WinEdt, Emacs, vi, TeXlipse, …

Online mode

- Overleaf
- ShareLaTeX

- Commands start with a backslash (“\”)
- Every document starts with a “\document class” command
  - \document class {<document type>} → LATEX what kind of document we are creating:
    - article
    - report
    - book
    - letter
- The body of the document must present between the \begin{document} and \end{document}
- Any text that comes after the \end{document} command will be ignored

Document class

article	For articles in scientific journals, presentations, short reports, program documentation, invitations, ...
IEEEtran	For articles with the IEEE Transactions format
proc	A class for proceedings based on the article class
report	For longer reports containing several chapters, small books, thesis, ...
book	For real books
slides	For slides
memoir	For changing sensibly the output of the document. It is based on the book class, but you can create any kind of document with it
letter	For writing letters
beamer	For writing presentations

```
\documentclass[options]{report}
```

Document class options

10pt, 11pt, 12pt	Sets the size of the main font in the document. Default: 10pt.
a4paper, letterpaper,...	Defines the paper size. Default: letterpaper.
fleqn	Typesets displayed formulas left-aligned instead of centered.
leqno	Places the numbering of formulas on the left hand side instead of the right.
twocolumn	Instructs LaTeX to typeset the document in two columns instead of one.

```
\documentclass[11pt,a4paper]{repo
rt}
```

Document Environment

Top matter

```
\documentclass[11pt,a4paper]{report}

\begin{document}
\title{How to Structure a LaTeX Document}
\author{Leslie Lamport}
\date{December 1984}
\maketitle
\end{document}
```

Sectioning Commands

Command	Level
\part{"part"}	-1
\chapter{"chapter"}	0
\section{"section"}	1
\subsection{"subsection"}	2
\subsubsection{"subsubsection"}	3
\paragraph{"paragraph"}	4
\subparagraph{"subparagraph"}	5

Itemize

- Used in LaTeX environment to create unordered lists

```
\begin{itemize}
        \item <Sentence1>
        \item <Sentence2>
        \item <Sentence3>
\end{itemize}
```

- Basic example of using itemize

```
\begin{itemize}
    \item Bullet point 1
    \item Bullet point 2
    \item Bullet point 3
\end{itemize}
```

Output:

- Bullet point 1
- Bullet point 2
- Bullet point 3

Itemize (1)

Change the labels using in itemize

- **\usepackage**[option1,option2,option3]{package_name}

```
\documentclass{article}
\usepackage{amssymb}  %  to access $\blacksquare$
\begin{document}
```

```
Changing labels:
\begin{itemize}
  \item Using normal label for itemize
  \item Using another normal label for itemize
  \item[!] A point to exclaim something!
  \item[$\blacksquare$] Using Black Square.
  \item[NOTE] This entry has no bullet
  \item[] A blank label
\end{itemize}
\end{document}
```

Output:

Changing labels:

- Using normal label for itemize
- Using another normal label for itemize

! A point to exclaim something!

■ Using Black Square.

NOTE This entry has no bullet

A blank label

enumerate

- Used in LaTeX environment to create ordered lists

```
\begin{enumerate}
        \item <Sentence1>
        \item <Sentence2>
        \item <Sentence3>
\end{enumerate}
```

- Basic example of using enumerate

```
\begin{enumerate}
   \item First item
   \item Second item
   \item Third item
\end{enumerate}
```

Output:

1. First item
2. Second item
3. Third item

enumerate (1)

- **Change the labels using in enumerate**

```
\documentclass{article}
\usepackage{amssymb}  ->  to access $\blacksquare$
\begin{document}


Changing labels:
\begin{enumerate}
  \item Using normal label for enumerate
  \item Using another normal label for enumerate
  \item[!] A point to exclaim something!
  \item[$\blacksquare$] Using Black Square.
  \item[NOTE] This entry has no bullet
  \item[] A blank label
\end{enumerate}
\end{document}
```

Output:

Changing labels:

1. Using normal label for enumerate
2. Using another normal label for enumerate

! A point to exclaim something!

■ Using Black Square.

NOTE This entry has no bullet

A blank label

Nested Lists using itemize and enumerate

```
\documentclass{article}
\begin{document}
\begin{enumerate}
  \item The labels consist of sequential numbers
  \begin{itemize}
    \item  Normal \emph{itemize} example
    \item The text in the entries may be of any length
    \begin{itemize}
    \item[Note:] I would like to describe something here
    \item[Wait!] And give a warning here
    \end{itemize}
  \end{itemize}
  \item  Normal \emph{enumerate} example
\end{enumerate}
\end{document}
```

Output:

1. The labels consist of sequential numbers
   - Normal *itemize* example
   - The text in the entries may be of any length

   Note: I would like to describe something here

   Wait! And give a warning here
2. Normal *enumerate* example

Note: \emph is used to emphasise the text usually by putting it in italics

LATEX: math

- LaTeX's features for typesetting mathematics make it a compelling choice for writing technical documents.
- LaTeX allows two writing modes for mathematical expressions:
  - Inline math mode → Used to write formulas that are part of a paragraph
  - Display math mode → Used to write expressions that are not part of a paragraph, and are therefore put on separate lines

To produce an inline expression, place the math expression between dollar signs (“$”).

hello.tex

```
\document class{article}
\begin{document}
Let $y=mx+b$ be \ldots
\end{document}
```

Output:

Let $y = mx + b$ be . . .

Output:

```
\document class{article}
\usepackage{amsmath}
\begin{document}
\[\sqrt{x^2+1}\]
\end{document}
```

$$\sqrt{x^2+1}$$

LATEX: math

Description	Command	Output
Plus or minu	\pm	±
Multiplication (times)	\times	*
Multiplication (dot)	\dot	.
Division symbol	\div	÷
Not equal	\ne	≠
Infinity	\infinity	∞
Less than or equal to	\le	≤
Greater than or equal to	\ge	≥
Dots	1, 2, 3, \ldots	1, 2, 3, …
Dots	1+2+3+\cdots	1 + 2 + 3 + …
Fraction	\frac{a}{b}	a/b

An Interactive Introduction to LaTeX

Part 1: The Basics

Dr John D. Lees-Miller



January 26, 2020



Why LaTeX?

- It makes beautiful documents
  - Especially mathematics
- It was created by scientists, for scientists
  - A large and active community
- It is powerful — you can extend it
  - Packages for papers, presentations, spreadsheets, . . .

How does it work?

- You write your document in plain text with commands that describe its structure and meaning.
- The latex program processes your text and commands to produce a beautifully formatted document.

```
The rain in Spain falls \emph{mainly} on the plain.
```



The rain in Spain falls *mainly* on the plain.

More examples of commands and their output. . .

```
\begin{itemize}
\item Tea
\item Milk
\item Biscuits
\end{itemize}
```

- Tea
- Milk
- Biscuits

```
\begin{figure}
\includegraphics{gerbil}
\end{figure}
```



```
\begin{equation}
\alpha + \beta + 1
\end{equation}
```

$$\alpha + \beta + 1 \qquad (1)$$

Attitude adjustment

- Use commands to describe ‘what it is’, not ‘how it looks’.
- Focus on your content.
- Let LaTeX do its job.

Getting started

- A minimal LaTeX document:

```
\documentclass{article}
\begin{document}
Hello World! % your content goes here...
\end{document}
```

  - Commands start with a *backslash* \
  - Every document starts with a \documentclass command.
  - The *argument* in curly braces { } tells LaTeX what kind of document we are creating: an article.
  - A percent sign % starts a *comment* — LaTeX will ignore the rest of the line.

Getting started with **Overleaf**

- Overleaf is a website for writing documents in LATEX.
- It ‘compiles’ your LATEX automatically to show you the results.

Click here to open the example document in **Overleaf**

For best results, please use Google Chrome or a recent FireFox.

- As we go through the following slides, try out the examples by typing them into the example document on Overleaf.
- **No really, you should try them out as we go!**

Typesetting Text

- Type your text between \begin{document} and \end{document}.
- For the most part, you can just type your text normally.

Words are separated by one or more spaces. Paragraphs are separated by one or more blank lines.	Words are separated by one or more spaces. Paragraphs are separated by one or more blank lines.

- Space in the source file is collapsed in the output.

The rain in Spain falls mainly on the plain.	The rain in Spain falls mainly on the plain.

Typesetting Text: Warning

- Quotation marks are a bit tricky:
  use a backtick ` on the left and an apostrophe ' on the right.

Single quotes: `text'. Double quotes: ``text''.	Single quotes: ‘text’. Double quotes: “text”.

- Some common characters have special meanings in LATEX:

%	percent sign
#	hash (pound / sharp) sign
&	ampersand
$	dollar sign

If you just type these, you’ll get an error. If you want one to appear in the output, you have to escape it by preceding it with a backslash.

\$\%\&\#!	$%&#!

Handling Errors

- LATEX can get confused when it is trying to compile your document. If it does, it stops with an error, which you must fix before it will produce any output.
- For example, if you misspell `\emph` as `\meph`, LATEX will stop with an "undefined control sequence" error, because "meph" is not one of the commands it knows.

Advice on Errors

1. Don't panic! Errors happen.
2. Fix them as soon as they arise — if what you just typed caused an error, you can start your debugging there.
3. If there are multiple errors, start with the first one — the cause may even be above it.

Typesetting Exercise 1

Typeset this in LATEX: [1]

In March 2006, Congress raised that ceiling an additional $0.79 trillion to $8.97 trillion, which is approximately 68% of GDP. As of October 4, 2008, the "Emergency Economic Stabilization Act of 2008" raised the current debt ceiling to $11.3 trillion.

Click to open this exercise in **Overleaf**

- Hint: watch out for characters with special meanings!
- Once you've tried, click here to see my solution.

Typesetting Mathematics: Dollar Signs

- Why are dollar signs $ special? We use them to mark mathematics in text.

```
% not so good:
Let a and b be distinct positive
integers, and let c = a - b + 1.
% much better:
Let $a$ and $b$ be distinct positive
integers, and let $c = a - b + 1$.
```

Let a and b be distinct positive integers, and let c = a - b + 1.

Let *a* and *b* be distinct positive integers, and let *c* = *a* – *b* + 1.

- Always use dollar signs in pairs — one to begin the mathematics, and one to end it.
- LaTeX handles spacing automatically; it ignores your spaces.

```
Let $y=mx+b$ be \ldots

Let $y = m  x  +  b$ be \ldots
```

Let *y* = *mx* + *b* be . . .

Let *y* = *mx* + *b* be . . .

Typesetting Mathematics: Notation

- Use caret ^ for superscripts and underscore _ for subscripts.

```
$y = c_2 x^2 + c_1 x + c_0$
```

$y = c_2 x^2 + c_1 x + c_0$

- Use curly braces { } to group superscripts and subscripts.

```
$F_n = F_n-1 + F_n-2$      % oops!

$F_n = F_{n-1} + F_{n-2}$ % ok!
```

$F_n = F_n - 1 + F_n - 2$

$F_n = F_{n-1} + F_{n-2}$

- There are commands for Greek letters and common notation.

```
$\mu = A e^{Q/RT}$

$\Omega = \sum_{k=1}^{n} \omega_k$
```

$\mu = Ae^{Q/RT}$

$\Omega = \sum_{k=1}^{n} \omega_k$

Typesetting Mathematics: Displayed Equations

- If it's big and scary, *display* it on its own line using \begin{equation} and \end{equation}.

```
The roots of a quadratic equation are given by
\begin{equation}
x = \frac{-b \pm \sqrt{b^2 - 4ac}}
                    {2a}
\end{equation}
where $a$, $b$ and $c$ are \ldots
```

The roots of a quadratic equation are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (2)$$

where a, b and c are . . .

Caution: LaTeX mostly ignores your spaces in mathematics, but it can't handle blank lines in equations — don't put blank lines in your mathematics.